HKY replay: four legacy caps and controlled stop

2026-10-26

The legacy batch stopped after four actual 900-second timeouts. Zero primary rank/verdict outputs completed, so the independent replay remains inconclusive. There is no independent solver-word count, verified A/A rank pair or minimum guess count from these executions. This report describes that stopped batch; the subsequent native continuation has a separate prospective registration.

Fixed workload and observations

Original protocol: `protocol-hky-restart-20261026.json`, frozen before launch at 19:50:38 UTC. It registered both real arms on all 124 fixed draw/guess/branch tuples: 248 children, single thread, 8 GiB address-space limit, 900 seconds per child and no timeout retries. The source/build executable, input and producer receipt were independently pinned. Fresh processes test determinism; all clocks are invalid for speed inference. Host: the recorded shared Linux workspace, without admitted performance isolation. The orchestration was historical Python/Bash around the unchanged native scientific executable.

Actual original tuple	hky	hky-aa
d0-k0-b0	timeout	timeout
d0-k1-b0	timeout	timeout

The native aggregate records 248 planned, four terminated, zero verified, four timeouts and 244 unexecuted. There are zero verified A/A rank/verdict pairs, no observed mismatches among such pairs, and all registered-prefix minima remain null. Zero checked pairs cannot establish agreement. Censored metrics and independent solver words remain absent. The original producer's one timeout and unknown cost are unchanged.

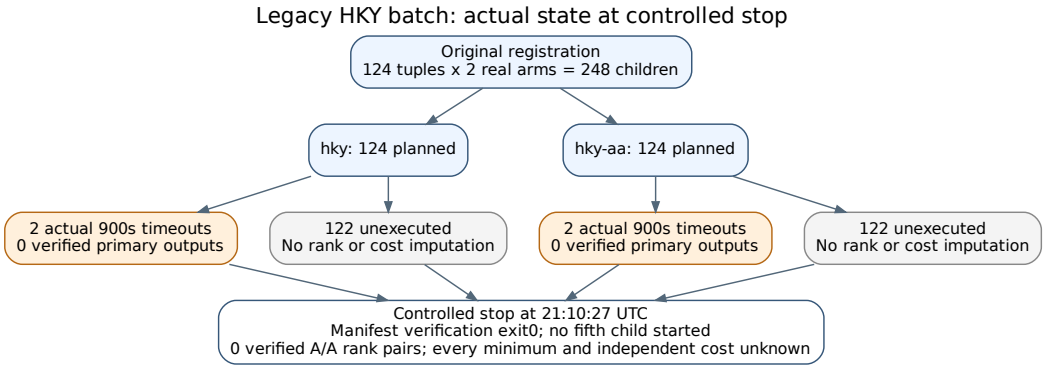


Figure 1: Counts at the controlled stop; unit: registered child processes, fixed census. There are no complete comparable primary metrics, so the diagram shows execution states rather than a performance comparison. Source: flow.dot, recorded-summary.json and the immutable legacy capsule.

Stop, provenance and restore

The controlled pause occurred at 21:10:27 UTC after the last completed case's checkpoint. No fifth case had a start marker. The recorded raw manifest check returned 0. The old batch launcher returned 143 for the intentional stop; the pause watcher reported outer exit1 despite its successful state files. Both observations are retained and neither is counted as an extra scientific failure. Reviewing restore ordering found PID/PGID metadata being rewritten before its old checksum check; the execution path also used prohibited Python orchestration. These are implementation defects, not mathematical obstructions.

The immutable legacy capsule is 18,132 bytes, SHA256:

`2f6d57ccdaf1738a8a1a7895559505a02a9f66abb948d09a221472ade95e812c`.

Use the repository artifact pull on `evidence/runs-hky-legacy-orchestration-20261026.tar.xz`, check the sibling SHA256 file from that directory and extract separately. The capsule retains the entire raw tree, original protocol/source and controlled-stop receipts. Its genuine binding and actual S3 GET receipt match the bytes. Original execution source is `e01cb6ebbf05`; the protocol's canonical digest is:

`32fbd8c1e4d6b6d1f7dcde2cd06084a28551ad4954922621193846b862277f31`.

The unchanged scientific executable is bound to source/build `a214e78d7`. The complete hashes are in the protocol and capsule. Its HKY/Kosters–Yeo literature, exact fixtures and bounded-degree limitations are recorded in PROTOCOL.md and COMPLETED_GUESS_REPLAY_PROTOCOL.md. This stop adds no degree, cost-scaling or full IC conclusion.

Continuation boundary

The new Rust controller verifies old manifests before PID/host writes and replaces the relevant Python execution path. Eleven tests, strict all-target lint and release build pass on its recorded source. A separate prospective 122-tuple, 244-child protocol froze before new data; legacy timeouts are never retried or imported as new prospective rows. The first actual native child started 22:35:44 UTC after quiet S3 checkpoint/archive/meta/resume/source-bundle readback. That launch is not a completed replay; see native transition and launch receipts.

The total additional-cloud ceiling remains USD100 including replay, no extra VM launched, and workspace billing is unmeasured. Current-head review/CI and remaining large-memory/full-price work remain open. Canonical cost and degree figures are unchanged: this censored result supplies no admitted new value.